NEUMANN HOT SPOTS ON ELLIPSOIDS IN ARBITRARY DIMENSION

Abstract. We give a proof of the strict hot-spots property for every bounded ellipsoid in $\mathbb{R}^n$, $n \geq 2$. More precisely, each nonzero eigenfunction for the first positive Neumann eigenvalue is strictly increasing in an appropriate direction in the longest-axis subspace. Its gradient therefore never vanishes in the interior. The multiplicity of this eigenvalue equals the number of longest semiaxes. The proof first orders the coordinate-odd eigenvalues by lifting them to Robin problems on one fixed ball in dimension $n + 2$. For distinct semiaxes, a finite-dimensional continuation constructs the positive ground state on a half ellipsoid in separated confocal form. The zeros of its separation polynomial determine the signs of the derivatives of its one-dimensional factors. Analytic extension handles the singular coordinate sets, and a limit argument treats coincident semiaxes. Orthogonal symmetries, together with injectivity of the mean gradient, then cover all linear combinations in the first eigenspace. No bound on eccentricity or dimension is imposed.

Draft status. This candidate proof was obtained by our team on October 18, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

Contents

1. Introduction and main theorem 1
1.1. Background and relation to earlier results 2
1.2. Main result 2
1.3. Outline of the proof 3
2. The first eigenspace of a general ellipsoid 3
2.1. Odd ground states and the mean gradient 4
2.2. The dimension-raising Robin transformation 5
2.3. Strict form comparison and consequences 7
3. Separation and monotonicity of the longest-axis odd ground state 7
3.1. Existence of positive separated factors 8
3.2. Confocal coordinates and a globally smooth eigenfunction 10
3.3. Strict monotonicity and the locations of the extrema 13
4. Coincident semiaxes and the full first eigenspace 14
References 16

1. Introduction and main theorem

Let $\Omega \subset \mathbb{R}^n$ be a bounded connected smooth domain. We number its Neumann eigenvalues by

$$0 = \mu_0(\Omega) < \mu_1(\Omega) \leq \mu_2(\Omega) \leq \cdots,$$

with multiplicities, and write

$$E_1(\Omega) = \{u : -\Delta u = \mu_1(\Omega) u \text{ in } \Omega,\ \partial_\nu u = 0 \text{ on } \partial\Omega\}.$$

All eigenfunctions considered below are real valued. The strict hot-spots property means that, for every $0 \neq u \in E_1(\Omega)$,

$$\min_{\partial\Omega} u < u(x) < \max_{\partial\Omega} u \qquad (x \in \Omega). \tag{1.1}$$

Date: October 18, 2026.

In particular, every global maximum and every global minimum belongs to the boundary. Absence of interior critical points is a stronger conclusion.

1.1. **Background and relation to earlier results.** Jerison and Nadirashvili proved the hot-spots property for convex planar domains with two axes of symmetry, including the case of a multiple first positive eigenvalue [3]. This already includes planar ellipses. Chen, Li, and Wang established directional monotonicity of the least axially odd Neumann eigenfunction on suitable rotationally symmetric domains and obtained hot-spots results for long domains in arbitrary dimension [1]. Kennedy and Rohleder developed a vector variational method for higher-dimensional lip domains and a related class of symmetric domains [4]. Their geometric conditions impose restrictions beyond coordinate-reflection symmetry.

The planar theorem in the September 2026 preprint [6], concerning smooth simply connected planar domains, provides further context. No statement from that preprint is used in the proof here. In higher dimensions, convexity alone does not suffice: de Dios Pont constructed convex counterexamples in all sufficiently large dimensions [2]. Thus an argument valid for ellipsoids must use their particular geometry.

An affine map taking an ellipsoid to a ball does not preserve the Euclidean Neumann eigenproblem. Indeed, for

$$E_a = \left\{ x \in \mathbb{R}^n : \sum_{j=1}^{n} \frac{x_j^2}{a_j^2} < 1 \right\}, \qquad a_j > 0, \tag{1.2}$$

the substitution $x = D_a y$, $D_a = \operatorname{diag}(a_1, \dots, a_n)$, gives

$$-\operatorname{div}(D_a^{-2}\nabla v) = \mu v \quad \text{in } B^n, \qquad y \cdot D_a^{-2}\nabla v = 0 \quad \text{on } S^{n-1}. \tag{1.3}$$

Both the differential operator and its conormal boundary condition have changed. We use this transformation for variational comparisons and limits, while constructing the monotone eigenfunction in confocal coordinates of the original ellipsoid.

1.2. **Main result.** Put

$$a_* = \max_j a_j, \qquad I_* = \{j : a_j = a_*\}, \qquad L_* = \operatorname{span}\{e_j : j \in I_*\}, \qquad m = |I_*|.$$

Here L_* is the longest-axis subspace; when $m > 1$, every unit vector in this subspace is a longest-axis direction.

Theorem 1.1 (Ellipsoids)**.** *Let $n \geq 2$ and let E_a be given by (1.2), with arbitrary positive semiaxes. Then the following assertions hold.*

(i) *The multiplicity of $\mu_1(E_a)$ is exactly m. In particular, it is simple if and only if the longest semiaxis is unique.*

(ii) *For every $0 \neq u \in E_1(E_a)$, there is a unit vector $e \in L_*$ such that*

$$\partial_e u(x) > 0 \qquad (x \in E_a). \tag{1.4}$$

(iii) *The strict hot-spots inequalities (1.1) hold. For the direction in (1.4),*

$$\max_{\overline{E_a}} u = u(a_* e), \qquad \min_{\overline{E_a}} u = u(-a_* e). \tag{1.5}$$

If all semiaxes are distinct, these two points are the unique global maximum and minimum, respectively.

The same conclusions, after the corresponding change of coordinates, hold for every translated and rotated ellipsoid.

The theorem covers balls, prolate and oblate spheroids, triaxial ellipsoids, and their analogues in every dimension. No semiaxis ratio is assumed to be close to one. The uniqueness assertion for boundary extrema in part (iii) is stated only for distinct semiaxes; strict exclusion of interior extrema holds without that restriction.

1.3. **Outline of the proof.** First, an integrated Bochner identity and the first Neumann Poincaré inequality show that $u \mapsto \int_{E_a} \nabla u$ is injective on $E_1(E_a)$. This observation, also used in the convex-body spectral arguments of Klartag [5, Section 4], shows that the first eigenspace is spanned by functions odd in exactly one principal coordinate and even in the others.

Second, write the positive ground state in the x_i-odd subspace as $u_i = x_i h_i$. Replacing x_i by the radius in three coordinates turns h_i into the positive ground state of a Robin problem in dimension $n+2$. After all these lifted problems are put on the same unit ball, their pairwise form differences are scalar multiples of the sum of two coordinate energies and the squared boundary L^2 norm. This orders the odd eigenvalues strictly according to the reciprocal squared semiaxes and identifies the entire first eigenspace.

Third, suppose the semiaxes are distinct. Confocal coordinates range in n ordered intervals. We construct a common separation polynomial by a continuation of n regular one-dimensional ground-state problems. A Vandermonde integral shows that the Jacobian determinant never vanishes. An explicit root argument gives compact bounds along the continuation, so it reaches the required coefficients. This construction produces the separated eigenfunction itself; it does not assume a completeness theorem for separated solutions.

Fourth, the $n-1$ Neumann–Neumann intervals account for all the roots of the separation polynomial. Integrating the scalar equations then shows that every confocal factor is strictly decreasing. On the positive half-domain, increasing the longest Cartesian coordinate decreases all confocal coordinates; oddness handles the negative half-domain. The resulting directional derivative is therefore positive. The separated product extends analytically across coincident confocal roots, and the strong maximum principle removes the remaining exceptional points.

Finally, smooth convergence of the simple coordinate-odd ground states handles coincident semiaxes. Orthogonal symmetries in L_*, together with the injectivity of the mean gradient, then give the result for every first eigenfunction, including arbitrary linear combinations when $m > 1$.

2. The first eigenspace of a general ellipsoid

For the ellipsoid E_a in (1.2), put

$$b_k = a_k^{-2}, \qquad A = \operatorname{diag}(b_1, \dots, b_n). \tag{2.1}$$

For the reflection $S_i(x_1, \dots, x_i, \dots, x_n) = (x_1, \dots, -x_i, \dots, x_n)$, put

$$\lambda_i(a) = \inf_{\substack{0 \neq v \in H^1(E_a) \\ v \circ S_i = -v}} \frac{\displaystyle\int_{E_a} |\nabla v|^2}{\displaystyle\int_{E_a} v^2}. \tag{2.2}$$

An odd function has mean zero, so $\lambda_i(a) \geq \mu_1(E_a)$.

Theorem 2.1 (Ordering by semiaxis and exact multiplicity)**.** *For every $i, j \in \{1, \dots, n\}$,*

$$a_i > a_j \implies \lambda_i(a) < \lambda_j(a), \qquad a_i = a_j \implies \lambda_i(a) = \lambda_j(a). \tag{2.3}$$

Let

$$a_* = \max_{1 \leq k \leq n} a_k, \qquad I_* = \{i : a_i = a_*\}, \qquad m = |I_*|.$$

Then

$$\mu_1(E_a) = \lambda_i(a) \quad (i \in I_*), \qquad E_1(E_a) = \bigoplus_{i \in I_*} \operatorname{span}\{u_i\}, \tag{2.4}$$

where u_i is the least x_i-odd eigenfunction, chosen positive on $E_a \cap \{x_i > 0\}$. In particular,

$$\dim E_1(E_a) = m.$$

A unique longest semiaxis therefore gives a simple first positive Neumann eigenvalue.

The proof uses a dimension-raising transformation which turns each coordinate-odd problem into a positive Robin ground-state problem. After all lifted problems are put on the same unit ball, their quadratic forms are ordered pointwise.

2.1. **Odd ground states and the mean gradient.**

Lemma 2.2 (The least coordinate-odd eigenfunction)**.** *The infimum in (2.2) is attained and is simple within the x_i-odd subspace. Its eigenfunction u_i can be chosen strictly positive on $E_a \cap \{x_i > 0\}$. It is even in every coordinate x_j with $j \neq i$, and it is smooth on $\overline{E_a}$.*

Proof. Write

$$E_a^i = E_a \cap \{x_i > 0\}, \qquad \Gamma_D = E_a \cap \{x_i = 0\}, \qquad V_i = \{v \in H^1(E_a^i) : \operatorname{Tr}_{\Gamma_D} v = 0\}.$$

Restriction identifies the x_i-odd subspace of $H^1(E_a)$ with V_i: the zero trace on the flat face makes odd reflection an H^1 extension, inverse to restriction. Both quadratic integrals are multiplied by two under this reflection. Thus the mixed Rayleigh quotient on V_i has infimum $\lambda_i(a)$, with Neumann data on the curved part of the boundary. The half-ellipsoid is bounded, connected and Lipschitz, so the direct method and compact embedding into $L^2(E_a^i)$ give a minimizer. The already established inequality $\lambda_i(a) \geq \mu_1(E_a) > 0$ gives positivity of its eigenvalue.

Replacing a minimizer by its absolute value preserves the quotient. Interior elliptic regularity and the strong maximum principle give a minimizer $v > 0$ in E_a^i. The absolute value of every nonzero lowest eigenfunction is likewise strictly positive. Such an eigenfunction cannot change sign, since continuity and connectedness would otherwise give an interior zero. A lowest eigenspace of dimension at least two would contain a nonzero function orthogonal to v, forcing a change of sign. Hence this eigenvalue is simple.

Let u_i be the odd reflection of v. For any $\varphi \in H^1(E_a)$, put $\varphi_o = (\varphi - \varphi \circ S_i)/2$. Its restriction to E_a^i lies in V_i, and the mixed weak equation gives

$$\int_{E_a} \nabla u_i \cdot \nabla \varphi = 2 \int_{E_a^i} \nabla v \cdot \nabla \varphi_o = 2\lambda_i(a) \int_{E_a^i} v \varphi_o = \lambda_i(a) \int_{E_a} u_i \varphi.$$

Thus u_i is a weak Neumann eigenfunction on the full ellipsoid. Its smooth boundary and Neumann elliptic regularity give $u_i \in C^\infty(\overline{E_a})$.

Reflection in any other coordinate preserves the mixed problem, positivity, and the L^2 normalization. Simplicity therefore makes u_i even in that coordinate. □

Lemma 2.3 (Injectivity of the mean gradient)**.** *Let $\Omega \subset \mathbb{R}^n$ be a bounded connected smooth domain whose outward second fundamental form* II *is positive definite. For $u \in E_1(\Omega)$,*

$$\frac{\mu_1(\Omega)}{|\Omega|} \left| \int_\Omega \nabla u \right|^2 \geq \int_{\partial\Omega} \mathrm{II}(\nabla_\tau u, \nabla_\tau u). \tag{2.5}$$

Consequently $J : u \mapsto \int_\Omega \nabla u$ is injective on $E_1(\Omega)$.

Proof. Write $\mu = \mu_1(\Omega)$. Differentiating the Neumann condition in a tangent direction T gives

$$D^2 u(\nu, T) = -\mathrm{II}(\nabla_\tau u, T).$$

Since ∇u is tangent on the boundary, integration of the Bochner identity gives

$$\int_\Omega |D^2 u|^2 = \mu \int_\Omega |\nabla u|^2 - \int_{\partial\Omega} \mathrm{II}(\nabla_\tau u, \nabla_\tau u). \tag{2.6}$$

On the other hand, the Neumann Poincaré inequality applied to the components $\partial_k u$ gives

$$\int_\Omega |D^2 u|^2 \geq \mu \int_\Omega |\nabla u|^2 - \frac{\mu}{|\Omega|} \left| \int_\Omega \nabla u \right|^2 .$$

Comparison proves (2.5).

If $J(u) = 0$, positivity of II implies $\nabla_\tau u = 0$ everywhere on the boundary. Together with Neumann data, this gives $\nabla u = 0$ there. Moreover, every componentwise Poincaré inequality is an equality: their nonnegative defects sum to zero by (2.6). Set $w_k = \partial_k u$. Since w_k has mean zero, equality in the Poincaré inequality gives

$$\int_\Omega \nabla w_k \cdot \nabla \varphi = \mu \int_\Omega w_k \varphi \qquad (\varphi \in H^1(\Omega)).$$

Indeed, variations of mean zero give this identity, and constants also satisfy it because $\int_\Omega w_k = 0$. The trace of w_k vanishes because $\nabla u = 0$ on the boundary. Consequently its zero extension $\widetilde{w}_k$ belongs to $H^1(\mathbb{R}^n)$, with gradient equal to the zero extension of ∇w_k. For $\psi \in C_c^\infty(\mathbb{R}^n)$, the displayed identity applied to $\psi|_\Omega$ gives

$$\int_{\mathbb{R}^n} \nabla \widetilde{w}_k \cdot \nabla \psi = \mu \int_{\mathbb{R}^n} \widetilde{w}_k \psi.$$

Thus $(-\Delta - \mu)\widetilde{w}_k = 0$ distributionally on $\mathbb{R}^n$. Elliptic regularity and unique continuation imply $\widetilde{w}_k = 0$, since it vanishes on the nonempty exterior of $\overline{\Omega}$. Hence every $\partial_k u = 0$, so u is constant. Its zero mean gives $u = 0$. □

For the ellipsoid, $\nu(x) = Ax/|Ax|$, and a tangent vector T satisfies

$$\mathrm{II}_x(T,T) = \frac{T^T A T}{|Ax|} > 0.$$

Thus Lemma 2.3 applies.

Proposition 2.4 (Reduction to the coordinate-odd sectors)**.** *For every ellipsoid,*

$$\mu_1(E_a) = \min_{1 \leq i \leq n} \lambda_i(a), \qquad E_1(E_a) = \bigoplus_{\{i : \lambda_i(a) = \mu_1(E_a)\}} \mathrm{span}\{u_i\}. \tag{2.7}$$

Proof. The coordinate reflections act on the finite-dimensional eigenspace as commuting self-adjoint involutions. Decompose that space into their joint eigenspaces. A joint eigenfunction v has a parity $\sigma_j \in \{1, -1\}$ in each coordinate.

The integral $\int_{E_a} \partial_k v$ can be nonzero only when $\sigma_k = -1$ and $\sigma_j = 1$ for every $j \neq k$. Indeed, differentiation reverses the parity in the differentiated coordinate and leaves every other parity unchanged; an integrand odd in any coordinate has zero integral. Thus every joint parity pattern other than exactly one odd coordinate has $J(v) = 0$, and Lemma 2.3 eliminates it.

If a nonzero first eigenfunction is odd precisely in coordinate i, its quotient gives $\lambda_i(a) \leq \mu_1(E_a)$. Together with the already established reverse inequality, this gives equality, and simplicity in Lemma 2.2 identifies this component with a multiple of u_i. Conversely, whenever $\lambda_i(a) = \mu_1(E_a)$, its eigenfunction u_i belongs to the first eigenspace. At least one parity component of a nonzero first eigenfunction is nonzero, proving the minimum formula and the direct-sum description. □

2.2. **The dimension-raising Robin transformation.** Set $N = n + 2$, let B^N be the unit ball, and for each i define

$$D_i = \mathrm{diag}(a_1, \ldots, a_n, a_i, a_i), \qquad C_i = D_i^{-2} = \mathrm{diag}(b_1, \ldots, b_n, b_i, b_i). \tag{2.8}$$

The lifted ellipsoid is $\widetilde{E}_i = D_i B^N$.

Lemma 2.5 (Exact Robin representation on a common ball)**.** *For every i,*

$$\lambda_i(a) = \inf_{0\neq h\in H^1(B^N)} \frac{\mathfrak{r}_i[h]}{\displaystyle\int_{B^N} h^2}, \tag{2.9}$$

where the form domain is the full space $H^1(B^N)$ and

$$\mathfrak{r}_i[h] = \int_{B^N} \left\{ \sum_{k=1}^{n} b_k |\partial_k h|^2 + b_i\big(|\partial_{n+1}h|^2 + |\partial_{n+2}h|^2\big) \right\} + b_i \int_{S^{N-1}} h^2. \tag{2.10}$$

Its ground state is simple and strictly positive on $\overline{B^N}$.

Proof. Write $t = x_i$ and denote the other coordinates by z. By Lemma 2.2, the function u_i is smooth and odd in t. Define

$$q(t,z) = \int_0^1 \partial_i u_i(st,z)\, ds.$$

Then $u_i = tq$, and q is smooth and even in t. For statements up to the boundary, first take a smooth extension of u_i and odd-symmetrize it. This yields a smooth even extension of q near the closed ellipsoid.

For $X = (x_1,\dots,x_n,\xi_1,\xi_2) \in \widetilde{E}_i$, set

$$r = (x_i^2 + \xi_1^2 + \xi_2^2)^{1/2}, \qquad H_i(X) = q(r,z).$$

The evenness of q makes this lift smooth at $r = 0$. One can check this without any analyticity assumption: for each integer M, the even Taylor expansion in t is a polynomial in t^2 plus a remainder of order t^{2M+2} with the corresponding derivative bounds. After $t = r$, the polynomial is smooth in (x_i,ξ_1,ξ_2) and the remainder has vanishing derivatives through any prescribed finite order if M is chosen sufficiently large. The same argument with the smooth extension just described proves smoothness up to the lifted boundary.

The equation for $u_i = tq$ gives, when $t > 0$,

$$q_{tt} + \frac{2}{t} q_t + \Delta_z q + \lambda_i q = 0.$$

The first two terms form the radial Laplacian in three dimensions. Consequently

$$-\Delta H_i = \lambda_i H_i \quad \text{in } \widetilde{E}_i. \tag{2.11}$$

Smoothness extends the equation across $r = 0$. Moreover $q(r,z) > 0$ for $r > 0$, because $u_i > 0$ on its positive half-domain. Thus $H_i \geq 0$, is nonzero, and is strictly positive throughout the lifted interior by the strong maximum principle.

On ∂E_a with $t > 0$, the identity $0 = \partial_\nu(tq)$ and $\nu_i = b_i t/|Ax|$ give

$$\partial_\nu q + \frac{b_i}{|Ax|} q = 0.$$

The outward normal on $\partial\widetilde{E}_i$ is $C_iX/|C_iX|$. Its radial and remaining components are

$$\nu_r = \frac{b_i r}{|C_iX|}, \qquad \nu_k = \frac{b_k x_k}{|C_iX|} \quad (k \neq i).$$

The preceding identity therefore lifts exactly to

$$\partial_\nu H_i + \frac{b_i}{|C_iX|} H_i = 0 \quad \text{on } \partial\widetilde{E}_i. \tag{2.12}$$

Continuity gives the same equation at the boundary points where $r = 0$. The Robin coefficient is positive and smooth on this boundary.

The Robin form

$$\int_{\widetilde{E}_i} |\nabla H|^2 + \int_{\partial \widetilde{E}_i} \frac{b_i}{|C_i X|} H^2$$

has a simple ground state $\Phi_i > 0$ in $\widetilde{E}_i$, by the same compactness and fixed-sign argument as in Lemma 2.2. Write κ_i for its eigenvalue. Testing the weak equations for H_i and Φ_i against each other gives

$$(\lambda_i(a) - \kappa_i) \int_{\widetilde{E}_i} H_i \Phi_i = 0.$$

The integral is positive, so $\lambda_i(a) = \kappa_i$ and H_i is the ground state of the full Robin form.

It remains to put the full Robin form on a common domain. Under $X = D_i y$, the volume and surface changes of variables are

$$dX = (\det D_i)\, dy, \qquad d\sigma_X = (\det D_i)|D_i^{-1} y|\, d\sigma_y \quad (y \in S^{N-1}).$$

Since $C_i D_i = D_i^{-1}$,

$$\frac{b_i}{|C_i X|} = \frac{b_i}{|D_i^{-1} y|}.$$

The surface Jacobian cancels this denominator, while the energy transforms by D_i^{-T}. The affine pullback is a bijection between $H^1(\widetilde{E}_i)$ and $H^1(B^N)$, so no symmetry restriction is placed on the transformed form domain. Dividing numerator and denominator of the Rayleigh quotient by $\det D_i$ gives exactly (2.10) and (2.9).

Finally, H_i is nonnegative on $\overline{\widetilde{E}_i}$ by continuity and satisfies $-\Delta H_i = \lambda_i(a) H_i \geq 0$. If it vanished at a boundary point, Hopf's boundary lemma on the smooth lifted ellipsoid would give $\partial_\nu H_i < 0$ there, contradicting (2.12). Thus $H_i > 0$ on $\overline{\widetilde{E}_i}$, and its pullback is strictly positive on $\overline{B^N}$ as claimed. □

2.3. **Strict form comparison and consequences.**

Proof of Theorem 2.1. Suppose $a_i > a_j$, equivalently $b_i < b_j$. The first n derivative coefficients in (2.10) are the same for both forms, and therefore

$$\mathfrak{r}_j[h] - \mathfrak{r}_i[h] = (b_j - b_i) \left\{ \int_{B^N} (|\partial_{n+1} h|^2 + |\partial_{n+2} h|^2) + \int_{S^{N-1}} h^2 \right\}. \tag{2.13}$$

Let h_j be the positive ground state of $\mathfrak{r}_j$, normalized in $L^2(B^N)$. Its boundary integral is strictly positive by Lemma 2.5. Testing the ith quotient with h_j gives

$$\lambda_i(a) \leq \mathfrak{r}_i[h_j] < \mathfrak{r}_j[h_j] = \lambda_j(a).$$

If $a_i = a_j$, then $b_i = b_j$ and the two forms are identical, so their ground eigenvalues are equal. This proves (2.3).

Proposition 2.4 and the strict ordering just proved give (2.4) and $\dim E_1(E_a) = |I_*|$. □

3. Separation and monotonicity of the longest-axis odd ground state

This section proves the monotonicity statement needed for the positive ground state on a half ellipsoid. In particular, the separated representation used below is obtained by an existence argument; it is not assumed from a formal separation ansatz.

First suppose the semiaxes are distinct. Set $c_i = a_i^2$, and permute the principal coordinates so that

$$0 < c_1 < c_2 < \cdots < c_n, \qquad E_a = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^n \frac{x_i^2}{c_i} < 1 \right\}. \tag{3.1}$$

Put

$$c_0 = 0, \qquad I_j = (c_{j-1}, c_j), \qquad P(t) = \prod_{i=1}^{n}(t - c_i), \qquad \sigma_j = \operatorname{sgn} P|_{I_j} = (-1)^{n-j+1}. \tag{3.2}$$

The intervals are transformed to finite regular intervals by

$$s_j(t) = \int_{c_{j-1}}^{t} \frac{\mathrm{d}r}{\sqrt{|P(r)|}}, \qquad S_j = s_j(c_j), \qquad t_j = s_j^{-1} : [0, S_j] \longrightarrow [c_{j-1}, c_j]. \tag{3.3}$$

All the numbers S_j are finite, since every zero of P is simple and $P(0) \neq 0$.

3.1. **Existence of positive separated factors.**

Lemma 3.1 (A regular one-dimensional continuation argument)**.** *There are a real polynomial Q of degree $n-1$ with positive leading coefficient μ, and positive functions f_j on $(0, S_j)$, such that*

$$-f_j''(s) + \frac{\sigma_j}{4} Q(t_j(s)) f_j(s) = 0. \tag{3.4}$$

For $j < n$ the endpoint conditions are Neumann at both ends. For $j = n$ they are Neumann at 0 and Dirichlet at S_n:

$$f_j'(0) = f_j'(S_j) = 0 \quad (j < n), \qquad f_n'(0) = 0, \quad f_n(S_n) = 0. \tag{3.5}$$

Every f_j is the positive principal eigenfunction for the indicated regular Sturm–Liouville problem, whose principal eigenvalue is zero. Moreover, Q has precisely one simple root in each interval I_j, $j < n$, and no other roots.

Proof. Choose points $\tau_j \in I_j$. For $0 \leq h \leq 1$ set

$$\theta_j^h(s) = (1-h)\tau_j + h t_j(s).$$

For $q = (q_0, \ldots, q_{n-1}) \in \mathbb{R}^n$, write $Q_q(t) = \sum_{\ell=0}^{n-1} q_\ell t^\ell$, and let $\nu_j(q, h)$ be the principal eigenvalue of

$$-\frac{\mathrm{d}^2}{\mathrm{d}s^2} + \frac{\sigma_j}{4} Q_q(\theta_j^h(s))$$

on $[0, S_j]$, with the endpoint conditions in (3.5). Regular one-dimensional spectral theory gives a simple principal eigenvalue and an eigenfunction positive in the open interval. The operator domains in $H^2(0, S_j)$ are fixed by the endpoint conditions, and the multiplication potentials depend smoothly on (q, h) in $L^\infty(0, S_j)$. Simplicity therefore gives locally continuously differentiable principal eigenvalues and normalized positive eigenfunctions.

At $h = 0$ the potentials are constant. Consequently, $\nu_j(q, 0) = 0$ for $j < n$ precisely when $Q_q(\tau_j) = 0$, while $\nu_n(q, 0) = 0$ precisely when

$$Q_q(\tau_n) = 4\left(\frac{\pi}{2S_n}\right)^2.$$

These n interpolation conditions have the unique solution

$$Q_q(t) = \frac{4(\pi/(2S_n))^2}{\prod_{j<n}(\tau_n - \tau_j)} \prod_{j<n}(t - \tau_j).$$

In particular, the set of parameters h for which all the ν_j vanish is nonempty.

We next verify local continuation. Normalize the positive principal eigenfunction of the jth problem by $\int_0^{S_j} f_j^2 \, \mathrm{d}s = 1$. Differentiating the normalized eigenvalue equation and pairing with f_j cancels the eigenfunction derivatives and gives

$$\frac{\partial \nu_j}{\partial q_\ell} = \frac{\sigma_j}{4} \int_0^{S_j} \theta_j^h(s)^\ell f_j(s)^2 \, \mathrm{d}s, \qquad 0 \leq \ell \leq n-1.$$

By multilinearity of the determinant, the determinant of this n by n Jacobian is

$$\frac{\prod_{j=1}^{n}\sigma_j}{4^n}\int_{\prod_j(0,S_j)}\prod_{i<j}(\theta_j^h(s_j)-\theta_i^h(s_i))\prod_{j=1}^{n}f_j(s_j)^2\,\mathrm{d}s_1\cdots\mathrm{d}s_n. \tag{3.6}$$

The integrand after the constant sign factor is strictly positive in the open product of intervals. Indeed, the ranges are ordered; at $h=1$ adjacent closed ranges meet only at an endpoint, and interior points remain strictly ordered. Thus the Jacobian is nonsingular, including at $h=1$. The implicit function theorem proves local continuation at every simultaneous zero.

It remains to prevent a solution from escaping to infinity in the coefficient variables. Consider any simultaneous zero with $0\le h\le 1$. For a Neumann–Neumann factor, integration of its differential equation gives

$$\int_0^{S_j} Q_q(\theta_j^h(s))f_j(s)\,\mathrm{d}s=0,\qquad j<n. \tag{3.7}$$

If $h>0$, positivity of f_j shows that Q_q has a root strictly inside the range of θ_j^h. To justify strictness, a nonzero polynomial with no interior root has one sign throughout that range, which is incompatible with (3.7). The polynomial cannot vanish identically: at $q=0$ the last problem has principal eigenvalue $(\pi/(2S_n))^2>0$. For $h=0$, (3.7) instead says $Q_q(\tau_j)=0$. The $n-1$ roots so obtained lie in the disjoint intervals $I_1,\dots,I_{n-1}$. Since $\deg Q_q\le n-1$, it follows that

$$Q_q(t)=\mu\prod_{j<n}(t-r_j),\qquad r_j\in[c_{j-1},c_j], \tag{3.8}$$

where all roots are distinct and simple. In fact each root is interior to the corresponding range if $h>0$.

Since $\sigma_n=-1$, the energy identity for the last factor is

$$\int_0^{S_n}|f_n'|^2\,\mathrm{d}s=\frac{\mu}{4}\int_0^{S_n}\prod_{j<n}(\theta_n^h(s)-r_j)f_n(s)^2\,\mathrm{d}s. \tag{3.9}$$

The product in the right side is positive in the open interval. The left side is positive because of the Dirichlet condition at S_n. Therefore $\mu>0$.

For a uniform upper bound, choose α with $c_{n-1}<\alpha<\tau_n$ and fix $0\ne\psi\in C_c^\infty((s_n(\alpha),S_n))$. This function is admissible for the Neumann–Dirichlet form. On its support, $\theta_n^h(s)>\alpha$ for every $0\le h\le 1$. With $\delta=\alpha-c_{n-1}>0$, all factors $\theta_n^h(s)-r_j$ are at least δ there. The variational characterization of $\nu_n=0$ yields

$$0\le\frac{\int|\psi'|^2\,\mathrm{d}s}{\int\psi^2\,\mathrm{d}s}-\frac{\mu}{4}\frac{\int\prod_{j<n}(\theta_n^h-r_j)\psi^2\,\mathrm{d}s}{\int\psi^2\,\mathrm{d}s}\le\frac{\int|\psi'|^2\,\mathrm{d}s}{\int\psi^2\,\mathrm{d}s}-\frac{\mu\delta^{n-1}}{4}.$$

Thus μ is uniformly bounded above. Equation (3.8) now bounds every coefficient of Q_q, uniformly over all simultaneous zeros and all $h\in[0,1]$.

The set of such h is relatively open by the implicit function theorem. To see that it is relatively closed, take simultaneous zeros $(q^{(k)},h_k)$ with $h_k\to h\in[0,1]$. The coefficient bounds give a subsequence $q^{(k)}\to q$. The corresponding potentials converge uniformly on each fixed interval $[0,S_j]$; the Rayleigh principle bounds the change of a principal eigenvalue by the uniform change of its potential. Hence $\nu_j(q,h)=0$ for every j. Since $[0,1]$ is connected, this set is all of $[0,1]$. At $h=1$ we obtain the asserted factors. The root argument at $h=1$ places their $n-1$ simple roots strictly inside $I_1,\dots,I_{n-1}$. □

Lemma 3.2 (Patching at singular endpoints)**.** *The factors furnished by Lemma 3.1 can be multiplied by positive constants so that*

$$F(t) = f_j(s_j(t)), \qquad t \in [c_{j-1}, c_j],$$

defines a single function on $[0, c_n]$*. This function is positive on* $[0, c_n)$*, is real analytic there including across* $c_1, \dots, c_{n-1}$*, and satisfies*

$$4P(t)F''(t) + 2P'(t)F'(t) = Q(t)F(t) \quad (0 \le t < c_n), \qquad F'(0) = 0. \tag{3.10}$$

Moreover,

$$F(t) = \sqrt{c_n - t}\, H(t), \tag{3.11}$$

where H *is positive and real analytic on a neighborhood of* $[0, c_n]$*.*

Proof. The chain rule and (3.3) transform (3.4) into (3.10) on every open interval I_j. We give the endpoint justification as well. At a simple zero c_i of P, fix one adjacent side and write $t = c_i + \varepsilon\rho^2$, where $\varepsilon \in \{-1, 1\}$ and $\rho \ge 0$. There is a real analytic function G_ε near zero such that

$$|P(c_i + \varepsilon\rho^2)| = \rho^2 G_\varepsilon(\rho^2), \qquad G_\varepsilon(0) = |P'(c_i)| > 0.$$

The distance d in the regular s coordinate from the endpoint is therefore

$$d = \int_0^\rho \frac{2\, \mathrm{d}\eta}{\sqrt{G_\varepsilon(\eta^2)}}.$$

This is odd and real analytic in ρ, with nonzero derivative at zero. Its analytic inverse $\rho = \rho(d)$ is also odd. Hence $t = c_i + \varepsilon\rho(d)^2$ is even and real analytic in d. The coefficient in (3.4) is consequently even and real analytic there. A solution with Neumann endpoint data is even and real analytic in d, by uniqueness for a regular ordinary differential equation. It is therefore analytic in d^2; the displayed integral makes d^2 analytic in ρ^2 and hence in $t - c_i$. Thus the corresponding F is real analytic in t at the endpoint. Its value is nonzero and positive, since a positive principal eigenfunction cannot have both its value and its derivative zero at a Neumann endpoint.

At an interior singular point c_i, the analytic solutions on the two sides agree after their positive endpoint values are matched. For completeness, uniqueness of the analytic branch can be seen directly by expanding at $t = c_i$. If $F = \sum_{k\ge 0} a_k (t - c_i)^k$, then, for every $k \ge 1$, the coefficient multiplying a_k in the coefficient equation of degree $k - 1$ is

$$2P'(c_i)k(2k - 1) \ne 0.$$

Thus all coefficients are uniquely determined by a_0. Successive positive rescalings patch the factors across $c_1, \dots, c_{n-1}$. At $t = 0$ the coordinate change is regular, so the first Neumann condition gives $F'(0) = 0$.

At c_n use $\varepsilon = -1$ and $\rho = \sqrt{c_n - t}$. The Dirichlet solution $f_n(S_n - d)$ is odd and real analytic in d, since the regular equation has an even analytic coefficient. Its zero is simple by uniqueness for the initial value problem, and its derivative into the interval is positive. Consequently $f_n(S_n - d(\rho))/\rho$ is even and real analytic in ρ, with positive value at zero. It is therefore analytic in $\rho^2 = c_n - t$, giving (3.11) near c_n. Away from c_n this quotient is positive and analytic by the already proved properties of F. The local extensions give the stated positive real analytic H on a neighborhood of the compact interval $[0, c_n]$. □

3.2. **Confocal coordinates and a globally smooth eigenfunction.** For $x \in E_a$ with all coordinates nonzero, let $\lambda_1(x), \dots, \lambda_n(x)$ be the roots of

$$\sum_{i=1}^n \frac{x_i^2}{c_i - t} = 1. \tag{3.12}$$

They satisfy

$$0 < \lambda_1 < c_1 < \lambda_2 < c_2 < \cdots < \lambda_n < c_n. \tag{3.13}$$

Indeed, the function $1+\sum_i x_i^2/(t-c_i)$ is strictly decreasing between successive poles, has opposite infinite limits at the two ends of each interval I_j with $j>1$, and is positive at 0 and negative near c_1 on I_1. Equivalently, the λ_j are the roots of the monic polynomial

$$R_x(t) = P(t) + \sum_{i=1}^{n} x_i^2 \frac{P(t)}{t-c_i} = \prod_{j=1}^{n}(t-\lambda_j). \tag{3.14}$$

Each quotient in the middle expression is a polynomial. In particular, R_x is defined without singularities when some x_i vanish. Evaluating at c_i gives the inverse relations

$$x_i^2 = \frac{\prod_j (c_i-\lambda_j)}{P'(c_i)}. \tag{3.15}$$

Conversely, for arbitrary $\lambda_j \in I_j$, the right side of (3.15) is positive: its numerator and denominator both have sign $(-1)^{n-i}$. Take the positive square roots to define x_i. The polynomial identity (3.14) then follows because its two monic sides agree at every c_i. Evaluating this identity at zero gives

$$1 - \sum_i \frac{x_i^2}{c_i} = \frac{\prod_j \lambda_j}{\prod_i c_i} > 0.$$

Thus the construction maps $\prod_j I_j$ into the positive orthant of E_a and is inverse to the root map. The roots are simple, so the implicit function theorem and the positive square-root formulas show that these maps are smooth inverses.

Lemma 3.3 (Metric and separated equation)**.** *On the positive orthant of E_a, the variables λ_j are orthogonal coordinates. If $\Delta_j = \prod_{k\neq j}(\lambda_j-\lambda_k)$, their metric coefficients are*

$$g_{jj} = -\frac{\Delta_j}{4P(\lambda_j)} > 0. \tag{3.16}$$

Consequently, for a twice differentiable function v,

$$\Delta v = -\sum_{j=1}^{n} \frac{4P(\lambda_j)v_{\lambda_j\lambda_j} + 2P'(\lambda_j)v_{\lambda_j}}{\Delta_j}. \tag{3.17}$$

In particular, $V(x) = \prod_j F(\lambda_j(x))$, with the factors of Lemma 3.2, satisfies $-\Delta V = \mu V$ at these regular coordinate points.

Proof. Put

$$B_j = \sum_{i=1}^{n} \frac{x_i^2}{(c_i-\lambda_j)^2} > 0.$$

Implicit differentiation of (3.12) gives

$$\frac{\partial \lambda_j}{\partial x_i} = -\frac{2x_i}{(c_i-\lambda_j)B_j}. \tag{3.18}$$

For $j\neq k$, subtracting the two equations (3.12) shows that $\sum_i x_i^2/((c_i-\lambda_j)(c_i-\lambda_k)) = 0$. Thus the gradients of distinct λ_j are orthogonal and $|\nabla\lambda_j|^2 = 4/B_j$. Differentiating (3.14) at a root gives $\Delta_j = -P(\lambda_j)B_j$, proving (3.16).

Let $\mathcal{V} = \prod_{i<j}(\lambda_j-\lambda_i) > 0$ and $p_j = \sqrt{|P(\lambda_j)|}$. The volume density is

$$\sqrt{\det g} = \frac{\mathcal{V}}{2^n \prod_j p_j}.$$

For fixed j, the quotient $\mathcal{V}/\Delta_j$ is independent of λ_j up to its constant sign. Therefore

$$\partial_{\lambda_j} \log(\sqrt{\det g}\, g^{jj}) = \frac{P'(\lambda_j)}{2P(\lambda_j)}.$$

Substitution into the coordinate formula $\Delta v = (\sqrt{\det g})^{-1} \sum_j \partial_{\lambda_j}(\sqrt{\det g}\, g^{jj} v_{\lambda_j})$ gives (3.17). For the product V, its right side divided by V is $-\sum_j Q(\lambda_j)/\Delta_j = -\mu$. The last equality is the leading-coefficient identity in Lagrange interpolation for a polynomial of degree at most $n-1$:

$$Q(t) = \sum_{j=1}^{n} Q(\lambda_j) \frac{\prod_{k\neq j}(t-\lambda_k)}{\Delta_j}.$$

□

The coordinate roots can collide when some Cartesian coordinates vanish. The following argument explicitly removes that apparent singularity from the separated eigenfunction.

Lemma 3.4 (Extension through colliding roots)**.** *There is a real analytic function U on a neighborhood of $\overline{E}_a$, odd in x_n and even in every other coordinate, which is positive on $E_a \cap \{x_n > 0\}$ and satisfies*

$$-\Delta U = \mu U \quad \text{in } E_a, \qquad \partial_\nu U = 0 \quad \text{on } \partial E_a. \tag{3.19}$$

At every regular point with $x_n > 0$ it is given, up to a fixed positive constant, by $\prod_j F(\lambda_j(x))$.

Proof. By continuity of polynomial roots, the ordered roots of R_x belong to $[c_{j-1}, c_j]$ for every $x \in \overline{E}_a$. From (3.15) with $i = n$,

$$\prod_j (c_n - \lambda_j) = x_n^2 P'(c_n), \qquad P'(c_n) > 0.$$

Thus, on regular points with $x_n > 0$,

$$\prod_j F(\lambda_j) = x_n \sqrt{P'(c_n)} \prod_j H(\lambda_j). \tag{3.20}$$

It suffices to show that the symmetric product on the right is real analytic even when roots collide.

Because H is positive and real analytic on $[0, c_n]$, it has a nonvanishing holomorphic extension to a sufficiently small simply connected complex neighborhood of that interval. Choose this neighborhood invariant under complex conjugation and let $\log H$ agree with the real logarithm on the interval. Then $\log H(\overline{z}) = \overline{\log H(z)}$. Choose a positively oriented simple contour Γ, surrounding $[0, c_n]$, whose closed interior lies in this neighborhood. Set

$$\mathcal{H}(x) = \exp\left\{ \frac{1}{2\pi \mathrm{i}} \int_\Gamma \log H(z) \frac{R_x'(z)}{R_x(z)}\, \mathrm{d}z \right\}.$$

All n zeros of R_x, counted with multiplicity, lie inside Γ for $x \in \overline{E}_a$. By compactness and continuity of polynomial roots, they remain inside and avoid Γ in a complex neighborhood of $\overline{E}_a$. The formula therefore defines a holomorphic function of x, since its integrand is holomorphic in x uniformly on Γ. The residue theorem, with multiplicities included, gives $\mathcal{H}(x) = \prod_j H(\lambda_j(x))$ on $\overline{E}_a$. For nearby real x, the roots occur in conjugate pairs, so the same residue formula and the symmetry of $\log H$ make $\mathcal{H}(x)$ real and positive. It is even in every x_i, since R_x depends only on the squares x_i^2.

Define $U(x) = x_n \mathcal{H}(x)$. This proves its stated parity, positivity and analyticity. Equation (3.20) and Lemma 3.3 prove the differential equation on the positive orthant. Reflection parity and invariance of the Laplacian extend it to every open orthant; analyticity then extends it to

all of E_a. At a point of ∂E_a with all Cartesian coordinates nonzero, the roots remain simple and extend smoothly to $\lambda_1 = 0$. Formula (3.18) gives

$$\nabla\lambda_1 = -\frac{2Ax}{|Ax|^2} \neq 0,$$

so this gradient is normal to the boundary. The orthogonality identities extend there by continuity; the product rule and $F'(0) = 0$ therefore give $\partial_\nu U = 0$. Such points are dense in ∂E_a, since each coordinate hyperplane has intersection of empty relative interior with the ellipsoidal boundary. Continuity of ∇U and of the normal extends the Neumann condition to the entire boundary. □

3.3. **Strict monotonicity and the locations of the extrema.** Write $E_a^+ = E_a \cap \{x_n > 0\}$. By Lemma 2.2, the restriction of the positive x_n-odd ground state is the mixed ground state on E_a^+.

Proposition 3.5 (The longest-axis mixed ground state)**.** *Under* (3.1)*, let u be the odd extension of the mixed ground state on E_a^+, with its sign chosen positive on E_a^+. Then*

$$\partial_{x_n} u(x) > 0 \qquad (x \in E_a). \tag{3.21}$$

Its unique global maximum on $\overline{E}_a$ is at $\sqrt{c_n}\, e_n$, and its unique global minimum is at $-\sqrt{c_n}\, e_n$.

Proof. Let U be furnished by Lemma 3.4. Both U and u are odd in x_n and positive on E_a^+, so $\int_{E_a} Uu > 0$. Orthogonality of Neumann eigenfunctions for distinct eigenvalues gives $\mu = \lambda_n(a)$, and simplicity in Lemma 2.2 identifies U with a positive multiple of u. Thus the separated representation holds for the actual ground state.

Write the roots of Q as $r_j \in I_j$, $j < n$. On I_j, $j < n$, their ordering and (3.2) give

$$\sigma_j Q(t) = \mu(t - r_j) \prod_{\substack{k<n \\ k\neq j}} |t - r_k|.$$

With $\rho_j = s_j(r_j)$, equation (3.4) therefore gives $f_j'' < 0$ on $(0, \rho_j)$ and $f_j'' > 0$ on (ρ_j, S_j). Integrating from the left Neumann endpoint before ρ_j, and backwards from the right Neumann endpoint after it, gives $f_j' < 0$ throughout $(0, S_j)$. On I_n the polynomial Q is positive and $\sigma_n = -1$, so $f_n'' < 0$ throughout the open interval; $f_n'(0) = 0$ implies $f_n' < 0$ there as well. The coordinate changes are increasing, and hence

$$F'(t) < 0 \quad (t \in I_j, 1 \le j \le n). \tag{3.22}$$

In particular F is strictly decreasing on the entire interval $[0, c_n]$.

At every point with $x_n > 0$ and all coordinates nonzero, all numbers $c_n - \lambda_j$ are positive. Equations (3.18) and (3.22) give

$$\partial_{x_n} u = u \sum_{j=1}^{n} \frac{F'(\lambda_j)}{F(\lambda_j)} \frac{\partial \lambda_j}{\partial x_n} > 0.$$

These points are dense in E_a^+. Continuity and evenness of $w = \partial_{x_n} u$ in x_n therefore give $w \ge 0$ throughout E_a. Finally w satisfies $-\Delta w = \mu w \ge 0$. If w vanished at an interior point, the strong minimum principle for the nonnegative superharmonic function w would force $w \equiv 0$. That is impossible because u is odd and positive on E_a^+. This proves (3.21), including at singular coordinate points.

For the extrema, put $p_+ = \sqrt{c_n}\, e_n$. Continuity of (3.20) gives the product representation on $\overline{E}_a \cap \{x_n \ge 0\}$. The ordered roots at p_+ are $(0, c_1, \dots, c_{n-1})$. Since $F(c_{j-1}) > 0$, we have

$$\frac{u(x)}{u(p_+)} = \prod_{j=1}^{n} \frac{F(\lambda_j(x))}{F(c_{j-1})} \le 1 \qquad (x \in \overline{E}_a,\ x_n \ge 0).$$

Every factor belongs to $[0,1]$. Strict decrease of F makes equality equivalent to $\lambda_j(x) = c_{j-1}$ for all j. Then $R_x(t) = t\prod_{i<n}(t-c_i)$, and (3.15) gives $x_i = 0$ for $i < n$ and $x_n^2 = c_n$. In the closed positive half ellipsoid this forces $x = p_+$. On the rest of $\overline{E}_a$ the function is nonpositive, whereas $u(p_+) > 0$. Thus p_+ is the unique global maximum, and oddness gives the unique global minimum at $-p_+$. □

4. Coincident semiaxes and the full first eigenspace

The preceding construction uses strict inequalities between the semiaxes. The coordinate-odd ground state, however, remains simple when axes coincide. We use this fact to pass to the limit before considering linear combinations in the first eigenspace.

Lemma 4.1 (Continuity of an odd ground state)**.** *Fix an index i and positive semiaxes a. If $a^{(k)} \to a$, let u_k and u be the positive-on-$x_i > 0$ ground states in the x_i-odd subspaces of $E_{a^{(k)}}$ and E_a. Normalize their pullbacks to B^n to have $L^2(B^n)$ norm one. Then*

$$v_k(y) := u_k(D_{a^{(k)}}y) \longrightarrow v(y) := u(D_a y) \quad \text{in } C^2(\overline{B^n}). \tag{4.1}$$

Their eigenvalues converge as well.

Proof. On the fixed odd subspace of $H^1(B^n)$ the quadratic forms are

$$q_k(f) = \int_{B^n} \nabla f \cdot A_k \nabla f, \qquad A_k = D_{a^{(k)}}^{-2} \longrightarrow A = D_a^{-2}.$$

Uniform ellipticity and convergence of the matrices give numbers $\varepsilon_k \downarrow 0$ such that

$$(1-\varepsilon_k)q(f) \le q_k(f) \le (1+\varepsilon_k)q(f), \qquad q(f) = \int_{B^n} \nabla f \cdot A \nabla f.$$

Taking infima of Rayleigh quotients proves convergence of the odd eigenvalues. The normalized v_k are bounded in $H^1(B^n)$. After passing to a subsequence, Rellich compactness gives $v_k \rightharpoonup v_*$ in H^1 and $v_k \to v_*$ in L^2. Thus v_* is odd, has norm one, and is nonnegative on $\{y_i > 0\}$, since the nonnegative cone is closed in L^2. Passing to the limit in the full weak eigenfunction equations gives

$$\int_{B^n} \nabla v_* \cdot A \nabla \varphi = \lambda_i(a) \int_{B^n} v_* \varphi \qquad (\varphi \in H^1(B^n)).$$

Testing with v_* shows that it attains the odd Rayleigh minimum. Simplicity and positivity from Lemma 2.2, together with its norm and sign, therefore give $v_* = v$.

To obtain convergence up to the boundary, apply (1.3) on the full ball:

$$-\operatorname{div}(A_k \nabla v_k) = \lambda_i(a^{(k)})v_k, \qquad y \cdot A_k \nabla v_k = 0 \quad \text{on } S^{n-1}.$$

The matrices range in a compact subset of the positive definite matrices; in particular the conormal direction satisfies $y \cdot A_k y \ge \kappa > 0$ on S^{n-1}, uniformly in k. The smooth coefficient norms of A_k and $y \mapsto A_k y$ are also uniformly bounded. The standard global elliptic estimates for this smooth conormal problem therefore give, for every nonnegative integer s,

$$\|v_k\|_{H^{s+2}(B^n)} \le C_s \left(\|\lambda_i(a^{(k)})v_k\|_{H^s(B^n)} + \|v_k\|_{L^2(B^n)} \right),$$

where C_s is independent of k. Iteration, starting with the L^2 normalization and the bounded eigenvalues, gives a bound in every fixed integer Sobolev norm. Choose an integer $M > n/2 + 2$; the compact embedding $H^M(B^n) \hookrightarrow C^2(\overline{B^n})$ then gives subsequential C^2 convergence. The already identified unique limit shows that the entire sequence converges as in (4.1). □

Proposition 4.2 (Monotonicity for every longest coordinate)**.** *For arbitrary positive semiaxes, suppose $a_i = a_*$. The positive x_i-odd ground state satisfies*

$$\partial_i u_i > 0 \quad \text{in } E_a. \tag{4.2}$$

Moreover,

$$\max_{\overline{E_a}} u_i = u_i(a_* e_i), \qquad \min_{\overline{E_a}} u_i = u_i(-a_* e_i). \tag{4.3}$$

Proof. Choose distinct positive numbers $d_1, \dots, d_n$ with $d_i > \max_{j \neq i} d_j$, and put

$$a_j^{(k)} = a_j + \varepsilon_k d_j,$$

where $\varepsilon_k \downarrow 0$ is sufficiently small to preserve every strict ordering among the original semiaxes. Equal original entries are separated by the distinct d_j. Thus every $a^{(k)}$ has distinct entries, converges to a, and has $a_i^{(k)}$ as its unique largest entry.

Both conclusions are invariant under positive rescaling. Normalize u_i and the positive x_i-odd ground states u_k on $E_{a^{(k)}}$ as in Lemma 4.1, and write

$$v_k(y) = u_k(D_{a^{(k)}} y), \qquad v(y) = u_i(D_a y).$$

After relabeling the coordinates, Proposition 3.5 gives $\partial_i u_k > 0$. Lemma 4.1 therefore gives

$$\partial_i u_i(D_a y) = \frac{1}{a_i} \partial_{y_i} v(y) = \lim_{k \to \infty} \frac{1}{a_i^{(k)}} \partial_{y_i} v_k(y) \geq 0 \quad (y \in B^n).$$

The function $w = \partial_i u_i$ is not identically zero, since $u_i(te_i) > u_i(0) = 0$ for $0 < t < a_i$. The strong minimum principle used in Proposition 3.5, applied to $-\Delta w = \lambda_i(a) w \geq 0$, now gives $w > 0$ throughout E_a.

The pole inequalities from that proposition are

$$v_k(-e_i) \leq v_k(y) \leq v_k(e_i) \qquad (y \in \overline{B^n}).$$

Uniform convergence gives the same inequalities for v. Since $D_a(\pm e_i) = \pm a_* e_i$, these are precisely (4.3). □

Proof of Theorem 1.1. Theorem 2.1 proves the multiplicity assertion and identifies the first eigenspace with the span of the u_i, $i \in I_*$. For $0 \neq u \in E_1(E_a)$ put

$$J(u) = \int_{E_a} \nabla u, \qquad e = \frac{J(u)}{|J(u)|}.$$

Lemma 2.3 makes the denominator nonzero, and the parity description shows that $J(u) \in L_*$. Fix $i \in I_*$ and normalize u_i to be positive on $x_i > 0$. Its parity and Proposition 4.2 imply

$$J(u_i) = c e_i, \qquad c = \int_{E_a} \partial_i u_i > 0.$$

Choose an orthogonal map R acting on L_* and fixing $L_*^\perp$ pointwise, with $Re_i = e$. When $m = 1$, this map acts by multiplication by 1 or -1 on L_*. Equal lengths of the maximal semiaxes imply $RE_a = E_a$. Thus $\widetilde{u}(x) = u_i(R^{-1}x)$ belongs to the first eigenspace and

$$J(\widetilde{u}) = \int_{E_a} R \nabla u_i(R^{-1}x) \, \mathrm{d}x = R \int_{E_a} \nabla u_i(y) \, \mathrm{d}y = ce,$$

where the change of variables uses $RE_a = E_a$ and $|\det R| = 1$. Injectivity of J now gives

$$u = \frac{|J(u)|}{c} \widetilde{u}.$$

Consequently

$$\partial_e u(x) = \frac{|J(u)|}{c} (\partial_i u_i)(R^{-1}x) > 0,$$

which proves (1.4). The pole values follow from (4.3) after applying R.

For an interior point x, let p_- and p_+ be the endpoints of its chord parallel to e, ordered in that direction. Strict monotonicity along the chord and continuity at its endpoints give

$$\min_{\partial E_a} u \le u(p_-) < u(x) < u(p_+) \le \max_{\partial E_a} u.$$

This proves (1.1), and $\partial_e u > 0$ also excludes interior critical points. For distinct semiaxes, the unique pole extrema in Proposition 3.5 transfer through R and the positive scaling. Finally, Euclidean rotations and translations preserve the Laplacian, the Neumann condition, and these conclusions. □

Remark 4.3 (The one-dimensional case). For the interval $(-a, a)$, the first positive Neumann eigenvalue is $\pi^2/(4a^2)$ and its eigenspace is spanned by $\sin(\pi x/(2a))$. The same strict hot-spots conclusion follows directly. The preceding argument is formulated for $n \ge 2$ so that the confocal interval construction and the curvature argument have their usual forms.

References

[1] H. Chen, Y. Li, and L. Wang, *Monotone properties of the eigenfunction of Neumann problems*, J. Math. Pures Appl. (9) **130** (2019), 112–129. doi:10.1016/j.matpur.2019.01.013.

[2] J. de Dios Pont, *Convex sets can have interior hot spots*, preprint, 2024. arXiv:2412.06344.

[3] D. Jerison and N. Nadirashvili, *The "hot spots" conjecture for domains with two axes of symmetry*, J. Amer. Math. Soc. **13** (2000), no. 4, 741–772. doi:10.1090/S0894-0347-00-00346-5.

[4] J. B. Kennedy and J. Rohleder, *On the hot spots conjecture in higher dimensions*, Trans. Amer. Math. Soc. Ser. B **13** (2026), 108–131. doi:10.1090/btran/245. Preprint: arXiv:2410.00816.

[5] B. Klartag, *A Berry–Esseen type inequality for convex bodies with an unconditional basis*, Probab. Theory Related Fields **145** (2009), nos. 1–2, 1–33. doi:10.1007/s00440-008-0158-6. Preprint: arXiv:0705.0832.

[6] OpenAI, *Strict hot spots and absence of interior critical points on smooth simply connected planar domains*, preprint, September 24, 2026. OpenAI mathematics repository.